

Faking Galaxies with Diffusion: A Real-vs-Synthetic Benchmark for Morphology Classification

Hunter Sandoval

Department of Electrical and Computer Engineering, University of New Mexico, Albuquerque, NM, USA

Abstract—This paper tests whether diffusion-generated galaxy images can augment or substitute for real training data in morphology classification. Stable Diffusion 1.5 is fine-tuned with LoRA on 12,330 Galaxy10 DECaLS images after removing two train/test leaks, and 134 ResNet-18 classifiers trained on real, synthetic and mixed data were evaluated once on 2,609 held-out real images, with the analysis plan fixed before the test set was read. Synthetic data provides a measurable benefit specifically when real data is scarce: with 10% of the real data, adding synthetic images raised accuracy by +2.2 points [1.2, 3.1], a gain that shrinks to zero as real data grows. It does not replace real data: synthetic-only training reaches 77% of real-only accuracy (66.9% vs. 86.6%), accuracy peaks with real data alone, augmenting all real data has no measurable effect, and up to 10% of the real data can be replaced at no detectable cost. The shortfall is substantially explained by class fidelity: 70% of generated images are recognized as their intended class versus 85% of real ones, and 28% for the rare Cigar-Shaped class.

Index Terms—diffusion models, synthetic data augmentation, galaxy morphology classification, generative models, transfer learning, low-rank adaptation.

I. INTRODUCTION

Labeled astronomical images are expensive, and some morphology classes are scarce: cleaned Galaxy10 DECaLS [1] has 333 Cigar-Shaped Smooth galaxies against 2,631 Round Smooth. Diffusion models now generate realistic galaxies [2], [3], but realistic is not the same as useful. This paper asks whether generated galaxies teach a classifier, combining latent diffusion [4] with LoRA [5], transfer learning (ResNet-18 [6]) and careful model evaluation [7], in a domain where scarcity is real.

The contribution is a leakage-controlled benchmark of class-conditional diffusion data for galaxy classification: (i) a fixed-size replacement curve with paired real-only controls, separating the value of synthetic images from the cost of less real data; (ii) generator settings chosen by downstream usefulness, not realism alone; (iii) a single test-set evaluation on a leakage-cleaned split, with the analysis plan fixed in advance; plus a rare-vs-common class analysis with a scarcity control. All code was written for this project and is public [8], with Claude Code (Anthropic) used as an AI-assisted development tool. Third-party components: the pretrained Stable Diffusion v1.5 and ResNet-18 weights, the libraries listed in Section V-E, the structure of the LoRA training loop (Hugging Face diffusers example *train_text_to_image_lora.py*, Apache-2.0) and the figure-palette checker, a Python port of a colour-blind-safety validator supplied with Claude Code's data-visualization guidance.

II. RELATED WORK

Table I compares the closest prior work. GalaxySD [3] already used diffusion galaxies to augment classifiers, with gains, so this paper does not claim to be the first such test; its novelty is the evaluation design and its mixed answer. Rows are not numerically comparable.

TABLE I. Related work and the new elements of this work

Method	Data; generator	Downstream evaluation	Reported finding	New in this work
GalCatDiff [2] (AJ, 2026)	Galaxies; diffusion, category embeddings	Realism only; no classifier trained	Realistic, physically consistent	Usefulness for a classifier, incl. rare class
GalaxySD [3] (ApJS, 2026)	Galaxy Zoo 2; conditional diffusion	Augmented classifiers; rare objects	Up to +30% completeness, purity	Replacement curve, paired controls, one test
Porcelain [9] (arXiv, 2026)	Porcelain; SD + LoRA	MobileNetV3; 95:5, 90:10 mixes	+3 to +5.5% macro-F1	Full 0-100% curve, scarcity control, per class
He et al. [10] (ICLR, 2023)	General; pretrained text-to-image	Few-shot; pre-training	Most useful when data is scarce	Fine-tuned generator, imbalanced data
Azizi et al. [11] (TMLR, 2023)	ImageNet; fine-tuned diffusion	Synthetic only (CAS); augmentation	CAS 64.96%; augmentation helps	Galaxies: 77% of real; no augm. gain
This work	Galaxy10 DECaLS; SD 1.5 + LoRA	ResNet-18; four arms; one test	+2.2 pts at 10% real; 66.9% vs. 86.6%	—

III. METHOD

A. Pipeline

Fig. 1 shows the full pipeline (pseudocode: Algorithm 1, Appendix I).

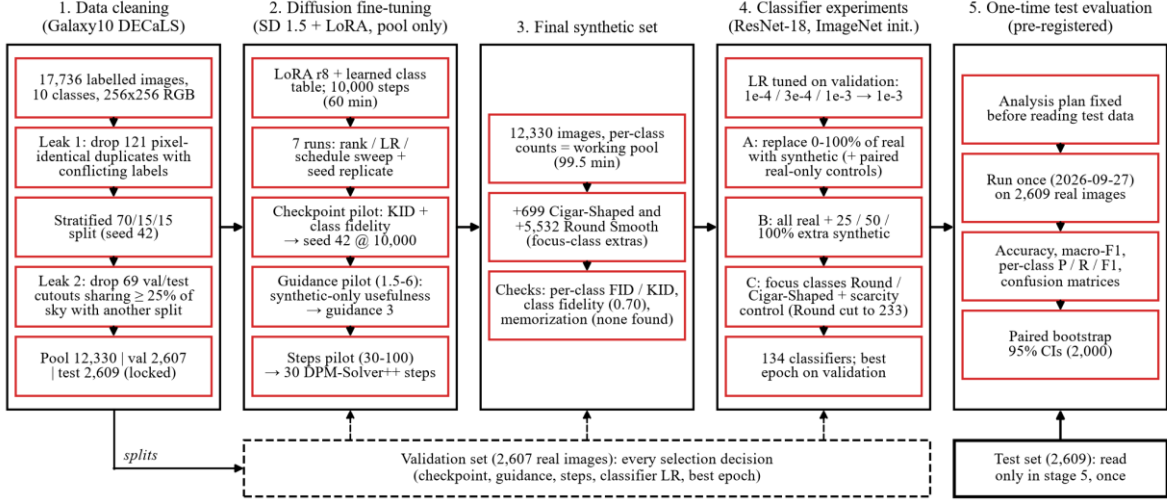


Fig. 1. Pipeline. Red boxes: steps; black boxes: stages. Every selection decision used the validation set (dashed); the test set was read once, in stage 5.

B. Models

Stable Diffusion v1.5 [4], pre-trained on LAION-5B subsets [12], has a VAE (83.7 M parameters; 512×512 images to 64×64×4 latents), a U-Net (859.5 M; residual blocks with self- and cross-attention) and a CLIP ViT-L/14 text encoder [13] (123.1 M), used here only to initialize an 11×77×768 class table (ten classes plus a null class for guidance [14]). Only LoRA (rank 8; 1.59 M parameters) and the table (0.65 M) are trained. *ResNet-18* [6] (18 weight layers, 11.18 M parameters, pre-trained on ImageNet-1k [15]) gets a new 10-class layer and is fully fine-tuned. Galaxies, small centered objects on dark sky, differ from both pretraining sets.

IV. DATA

A. Source

Galaxy10 DECaLS [1] (<https://github.com/henrysky/Galaxy10>; loaded with astroNN) has 17,736 RGB 256×256 cutouts from the DESI Legacy Surveys [16], labeled by Galaxy Zoo DECaLS volunteers [17] into ten classes of 334-2,645 images; no new data were collected. Fig. 2 shows their variation in size, orientation, brightness, color and neighbors (rows: Disturbed, Merging, Round Smooth, In-between Round, Cigar-Shaped Smooth, Barred Spiral, Unbarred Tight Spiral, Unbarred Loose Spiral, Edge-on w/o Bulge, Edge-on w/ Bulge).

B. No overlap between splits: two leaks removed

(1) 121 pixel-identical copies of 60 galaxies with conflicting labels, 30 of which a plain split had spread across splits, were removed before splitting. (2) Sky coordinates revealed 64 cross-split pairs of *different* catalogue entries whose cutouts share ≥25% of their sky (mostly Merging pairs); 69 validation/test images were removed after splitting, leaving the working pool and generators unchanged. No image is now shared between splits by index, pixels or sky.

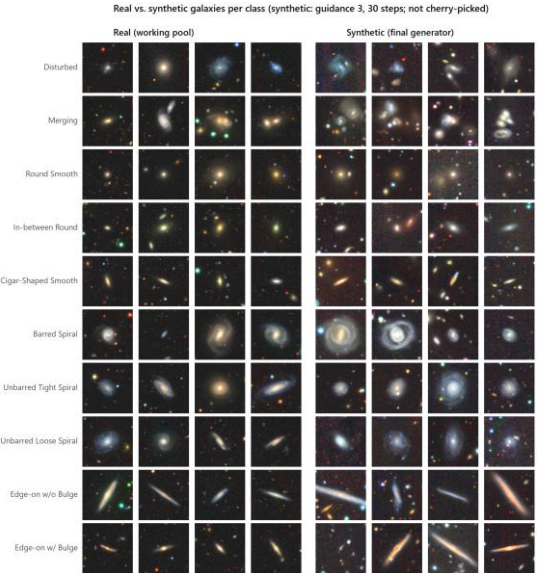


Fig. 2. Real (left) and synthetic (right; not cherry-picked) galaxies per class.

C. Training, validation and test sets

Stratified sampling (seed 42) gives a working pool for model fitting (12,330; 70%), a validation set for all selection decisions (2,607) and a test set used once (2,609) (Table II; per class: Appendix G). Synthetic images appear only in training sets.

TABLE II. Datasets for training (model fitting), validation and testing

	Galaxy10 DECaLS (real, cleaned)			Synthetic: main set			Synthetic: focus extras		
	All	Cigar	Round	All	Cigar	Round	All	Cigar	Round
Training	12,330	233	1,844	12,330	233	1,844	6,231	699	5,532
Validation	2,607	50	391	—	—	—	—	—	—
Testing	2,609	50	396	—	—	—	—	—	—
Total	17,546	333	2,631	12,330	233	1,844	6,231	699	5,532

Cigar = Cigar-Shaped Smooth (rare); Round = Round Smooth (common). The focus extras serve only experiment C.

V. EXPERIMENTAL SETUP

Models were fitted on the working pool and/or synthetic images; every choice used validation data. The test set was read once (2026-09-27), after the analysis plan was fixed, including the rule that among about 20 comparisons an isolated interval excluding zero is not a finding. Experiments: (A) *replacement*, 12,330 images with 0-100% synthetic at fixed per-class counts, plus paired real-only controls on each mix's real images; (B) *augmentation*, all real images plus 25-100% synthetic; (C) *focus classes*, 0.5-4× extra synthetic Round Smooth (1,844 real) or Cigar-Shaped (233 real) images, plus Round Smooth cut to 233 as a scarcity control. Repeated seeds on one split replace k-fold cross-validation, which would need a new generator per fold (160 min each): 3 seeds per condition (8 for real-only, Cigar-Shaped and the scarcity control), each redrawing the images. 95% CIs are paired bootstraps over test images (2,000 resamples).

A. Hyperparameter tuning

All tuning used only the validation split (Appendix A). Seven LoRA runs (rank 4-16, learning rate 5×10^{-5} to 2×10^{-4} , two schedules, two seeds) differed no more than two identically configured runs (KID 55.0 vs. 81.8), so rank 8 and 10^{-4} were kept, with the checkpoint best on KID and class fidelity. Guidance 3 and 30 steps were chosen by usefulness, the real-validation accuracy of a ResNet-18 trained only on 1,000 generated images (58.8% vs. at most 53.8% for other guidance values; 30 steps as useful as 100). The classifier learning rate (10^{-3}) and best epoch were chosen on validation too.

B. Training objectives

The generator minimizes the noise-prediction mean squared error of latent diffusion [18], [4] (with the class swapped for the null class 10% of the time, for guidance); minimizing it teaches the model each class's image distribution. Its value depends mostly on the random timestep and stayed at 0.193-0.203 while samples improved, so it has no useful target value; KID and class fidelity guided selection instead. The classifier minimizes cross-entropy (chance $\ln 10 \approx 2.30$; perfect 0). The multi-objective step is generator selection: raising guidance from 1.5 to 6 raised fidelity from 0.44 to 0.87 but worsened KID from 56.6 to 70.4, and usefulness, which was optimized directly, peaked at 3.

C. Optimization and compute

Both networks use AdamW [19], taken from the diffusers example; alternatives were not compared. Generator: learning rate 10^{-4} , weight decay 0.01, 500 warm-up steps then constant, batch 8, 10,000 steps (6.5 epochs), 60 min. Classifier: 10^{-3} , 0.05, one warm-up epoch then cosine, batch 128, 30 epochs, 1.5 min. On one NVIDIA RTX 5090 (32 GB), the generator (945.4 M parameters, 2.24 M trained) takes 920 ms and 50.8 TFLOPs per image and the classifier (11.18 M) 1.6 ms and 4.74 GFLOPs (Table XVI, Appendix H).

D. Training behavior

Classifier training loss fell from about 1.4 after one epoch to 0.02-0.06. The real-only and 50% models generalize well (validation loss 0.51 and 0.59; test within 0.8 points of validation in every replacement condition). The synthetic-only model reaches 99.6% training accuracy, but its validation loss is lowest at epoch 5 (1.42) and ends at 1.95: it transfers only partly to real galaxies. Training 50 instead of 30 epochs changed the real-vs-synthetic gap by 1 point (Appendix E). Out-of-distribution testing is discussed in Section VII.

E. Implementation

Python 3.14, PyTorch 2.14, diffusers 0.40, peft 0.21, torchvision 0.29, torchmetrics 1.9, torch-fidelity 0.4, astroNN 1.1 and scikit-learn 1.9. The public repository [8] (MIT license) holds the pipeline, all figures and tables with their data, six Colab tutorials, and a release with the weights, synthetic set and training logs (SHA-256 checksums).

VI. RESULTS

Metrics: accuracy, macro-F1 (weighting the 50-image rare class like the largest), per-class precision/recall/F1, confusion matrices and one-vs-rest ROC/AUC (on validation, because the one-time test saved classes, not scores; Appendix F). *Generator:* per-class FID [20] and KID [21] against real validation images; class fidelity, the share of generated images a real-trained ResNet-18 assigns to their class; and memorization (nearest training image in Inception features).

Synthetic data provides a measurable benefit specifically when real data is scarce (Fig. 3). Holding the real images fixed, adding synthetic images to 10% of the real data (1,233 images) raised test accuracy from 77.0% to 79.2%, a gain of +2.2 points [1.2, 3.1]. The interval excludes zero, and the gain shrinks steadily as real data grows (+0.6 [−0.2, +1.4] at 25%; −0.2 to +0.2 at 50-90%), the consistent pattern the analysis plan required for a claim. The benefit is bounded: 10% real plus synthetic stays below 25% real alone (81.9%), and the generator was trained on the full real pool, so this is the gain for a classifier with scarce real data and a generator built from plentiful data. Single classes showed no consistent effect (Cigar-Shaped F1 +0.00 to +0.02, all intervals include 0; Appendix D).

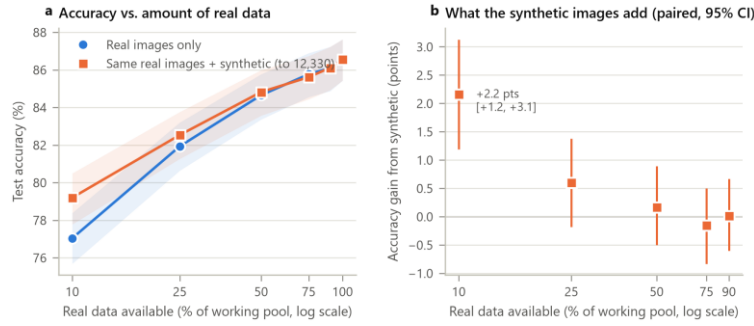


Fig. 3. Paired control: (a) test accuracy with only the real images of each mix vs. the same real images plus synthetic ones; (b) the paired gain from the synthetic images (95% CIs).

Synthetic data does not substitute for real data (Fig. 4): synthetic-only reaches 77.3% of real accuracy (a 19.6-point gap [18.1, 21.2]) and 73.7% of real macro-F1. Accuracy peaks with real data only. Up to 10% of the real data can be replaced at no detectable cost (−0.4 [−1.0, +0.1]); losses become significant from 25% and steepen past 75%. Adding synthetic images to all real data has no measurable effect (+0.3 [−0.3, +0.8]).

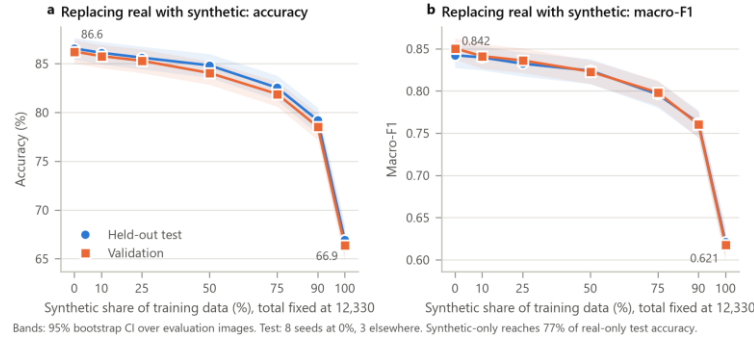


Fig. 4. Replacement curve: accuracy (a) and macro-F1 (b) as synthetic images replace real ones at a fixed total of 12,330 (bands: 95% CIs).

The gap is substantially explained by class fidelity, and it is worst for the rare class. A real-trained classifier recognizes 70% of generated images as their intended class versus 85% of real ones, so about 30% of synthetic images, twice the real rate, act like mislabeled examples. Fidelity is lowest for Cigar-Shaped (28%; real 84%) and Disturbed (29%; real 55%), the two classes that lose most: synthetic-only F1 is 0.375 vs. 0.769 and 0.287 vs. 0.557 (49% and 52% of real-only F1, against 69-92% for the others). Fig. 5 shows how: synthetic-only training sends 34% of real Cigar-Shaped galaxies to In-between Round, 31% of Disturbed to Unbarred Loose Spiral, and 36% of bulgeless edge-on galaxies to edge-on with bulge. Fidelity is not the whole story: past guidance 3 usefulness fell as realism worsened, and the final set's FID is 60.0 against a real reference of 20.0. No sign of memorization was found (Appendix B).

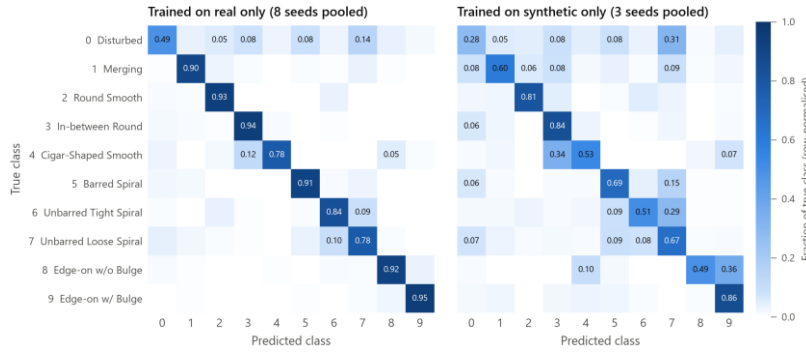


Fig. 5. Row-normalized test confusion matrices, real-only (8 seeds pooled) and synthetic-only (3 seeds). 0 Disturbed, 1 Merging, 2 Round Smooth, 3 In-between Round, 4 Cigar-Shaped, 5 Barred Spiral, 6 Unbarred Tight Spiral, 7 Unbarred Loose Spiral, 8 Edge-on w/o Bulge, 9 Edge-on w/ Bulge.

TABLE III. Classifier performance by training data (rows) and evaluation set (column groups)

Training data (12,330 images unless noted)	Held-out test set (2,609 real)			Validation set (2,607 real)		Seeds
	Accuracy % [95% CI]	Macro-F1	Δ acc. vs. real (pts)	Accuracy %	Macro AUC	
Real only	86.6 [85.4, 87.6]	0.842	—	86.2	0.983	8
10% synthetic	86.1 [84.9, 87.2]	0.840	-0.4 [-1.0, +0.1]	85.8	—	3
25% synthetic	85.6 [84.4, 86.7]	0.832	-0.9 [-1.5, -0.4]	85.3	—	3
50% synthetic	84.8 [83.6, 86.0]	0.824	-1.7 [-2.4, -1.1]	84.0	0.978	3
75% synthetic	82.5 [81.3, 83.8]	0.796	-4.0 [-4.9, -3.2]	81.9	—	3
90% synthetic	79.2 [77.8, 80.5]	0.762	-7.4 [-8.4, -6.3]	78.5	—	3
Synthetic only	66.9 [65.3, 68.5]	0.621	-19.6 [-21.2, -18.1]	66.4	0.927	3
All real + 100% synthetic (24,660)	86.8 [85.6, 88.0]	0.843	+0.3 [-0.3, +0.8]	86.3	—	3
10% real only (paired control, ~1,233)	77.0 [75.7, 78.4]	—	—	—	—	3

Same ResNet-18 (11.18 M parameters, 1.7 ms per image) in every row. Δ : paired bootstrap difference. AUC exists only for the three conditions whose validation scores were saved.

In Table III, accuracy and macro-F1 rank the models identically; AUC degrades less (0.927 vs. 0.983; at least 0.90 for 8 of 10 classes, 0.780 for Disturbed), so the synthetic-only model still ranks most classes but misplaces the boundaries. Augmentation doubles training cost for no gain.

VII. DISCUSSION AND LIMITATIONS

Synthetic galaxies from this generator give a measurable benefit when real data is scarce (+2.2 points with 10% of the real data) but cannot replace real data: alone they reach 77% of real accuracy, the best mix is all real, and up to 10% of the real data can be replaced at no detectable cost. The limit is class fidelity, lowest for the rare class that motivated the study; raising it (stronger conditioning, or filtering generated images with a real-trained classifier) is the main next step.

Limitations and future work. The evaluation is itself a train/test distribution shift: synthetic-only and mixed classifiers are trained wholly or partly on generated images and evaluated entirely on real, unseen galaxies. This is the idea of out-of-distribution testing, with synthetic data as the shifted domain instead of a different real-world source, and it is a deliberate design choice: holding the survey fixed isolates the training-data source as the one variable of interest. Whether the real-vs-synthetic gap also holds across real surveys, for example on Galaxy10 SDSS, is a natural extension left for future work, as is a stricter scarcity test in which the generator, too, sees only the scarce real data.

REFERENCES

- [1] H. W. Leung and J. Bovy, “Galaxy10 DECaLS,” 2019. [Dataset]. Available: <https://github.com/henrysky/Galaxy10>
- [2] X. Fan, H. Tang, Y. Zeng, M. B. N. Kouwenhoven, and G. Zeng, “Category-based galaxy image generation via diffusion models,” *Astron. J.*, vol. 171, no. 5, p. 268, Apr. 2026.
- [3] C. Ma *et al.*, “Can AI dream of unseen galaxies? Conditional diffusion model for galaxy morphology augmentation,” *Astrophys. J. Suppl. Ser.*, vol. 282, no. 2, p. 25, Jan. 2026.
- [4] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, “High-resolution image synthesis with latent diffusion models,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10674-10685.
- [5] E. J. Hu *et al.*, “LoRA: Low-rank adaptation of large language models,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.
- [6] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770-778.
- [7] S. Raschka, “Model evaluation, model selection, and algorithm selection in machine learning,” 2018, *arXiv:1811.12808*.
- [8] H. Sandoval, “Faking galaxies with diffusion: Code, models and data,” 2026. [Online]. Available: <https://github.com/HunterSandoval12/faking-galaxies-with-diffusion>

- [9] Z. Ling, S. Mirri, P. Salomoni, and G. Delnevo, "Synthetic data augmentation for multi-task Chinese porcelain classification: A Stable Diffusion approach," Jan. 2026, *arXiv:2601.14791*.
- [10] R. He *et al.*, "Is synthetic data from generative models ready for image recognition?" in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023.
- [11] S. Azizi, S. Kornblith, C. Saharia, M. Norouzi, and D. J. Fleet, "Synthetic data from diffusion models improves ImageNet classification," *Trans. Mach. Learn. Res.*, 2023.
- [12] C. Schuhmann *et al.*, "LAION-5B: An open large-scale dataset for training next generation image-text models," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS) Datasets and Benchmarks Track*, 2022.
- [13] A. Radford *et al.*, "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2021, pp. 8748-8763.
- [14] J. Ho and T. Salimans, "Classifier-free diffusion guidance," in *NeurIPS Workshop Deep Generative Models Downstream Appl.*, 2021, *arXiv:2207.12598*.
- [15] O. Russakovsky *et al.*, "ImageNet large scale visual recognition challenge," *Int. J. Comput. Vis.*, vol. 115, no. 3, pp. 211-252, 2015.
- [16] A. Dey *et al.*, "Overview of the DESI Legacy Imaging Surveys," *Astron. J.*, vol. 157, no. 5, p. 168, 2019.
- [17] M. Walmsley *et al.*, "Galaxy Zoo DECaLS: Detailed visual morphology measurements from volunteers and deep learning for 314,000 galaxies," *Mon. Not. R. Astron. Soc.*, vol. 509, no. 3, pp. 3966-3988, 2022.
- [18] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2020, pp. 6840-6851.
- [19] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019.
- [20] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs trained by a two time-scale update rule converge to a local Nash equilibrium," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2017, pp. 6626-6637.
- [21] M. Bińkowski, D. J. Sutherland, M. Arbel, and A. Gretton, "Demystifying MMD GANs," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2018.
- [22] C. Lu, Y. Zhou, F. Bao, J. Chen, C. Li, and J. Zhu, "DPM-Solver++: Fast solver for guided sampling of diffusion probabilistic models," 2022, *arXiv:2211.01095*.

APPENDIX

The appendices hold the pilot details, per-class breakdowns, the robustness check, ROC curves and the full result tables. All values are copied from the repository's results/ folder [8].

A. Generator training, sweep and sampling pilots

The diffusion loss stayed flat while sample quality changed, and generator quality swung strongly between checkpoints and between two identical runs with different seeds (Fig. 6, Table IV). Rank, learning rate and schedule made no difference beyond that noise (Fig. 7). The three candidate generators were compared at three guidance values (Table V), and guidance and steps were set by the usefulness pilots (Figs. 8, 9; Table VI).

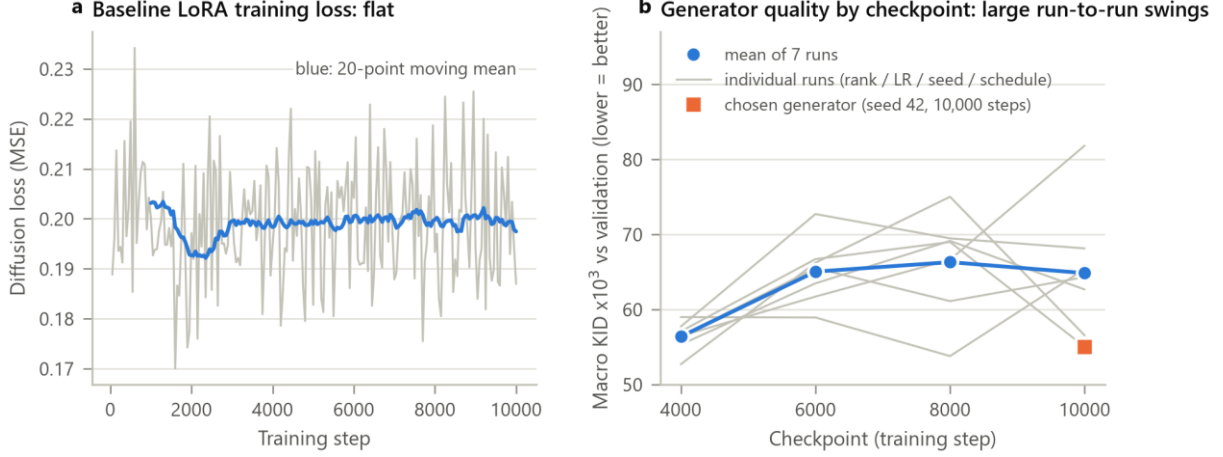


Fig. 6. (a) LoRA training loss (flat); (b) generator KID by checkpoint for all seven LoRA runs, chosen generator marked.

TABLE IV. LoRA sweep: macro KID x1e3 vs validation by checkpoint (guidance 2, 30 steps)

Run	2,000	3,000	4,000	5,000	6,000	8,000	10,000
baseline (rank 8, LR 1e-4, seed 42)	62.0	73.7	57.1	68.1	66.7	69.0	55.0
baseline, seed 43	75.6	59.8	56.5	69.0	61.7	66.6	81.8
rank 4	-	-	56.5	-	65.5	61.1	64.3
rank 16	-	-	59.0	-	58.9	53.8	65.4
LR 5e-5	-	-	55.3	-	63.5	69.1	62.7
LR 2e-4	-	-	52.7	-	66.2	75.0	56.5
cosine LR schedule	-	-	57.7	-	72.7	69.5	68.1

Lower is better. '-' = not scored. Same-settings seeds differ by up to 26.8 at 10,000 steps; no rank/LR setting beats that noise. Checkpoint 4,000 is the most consistent across runs.

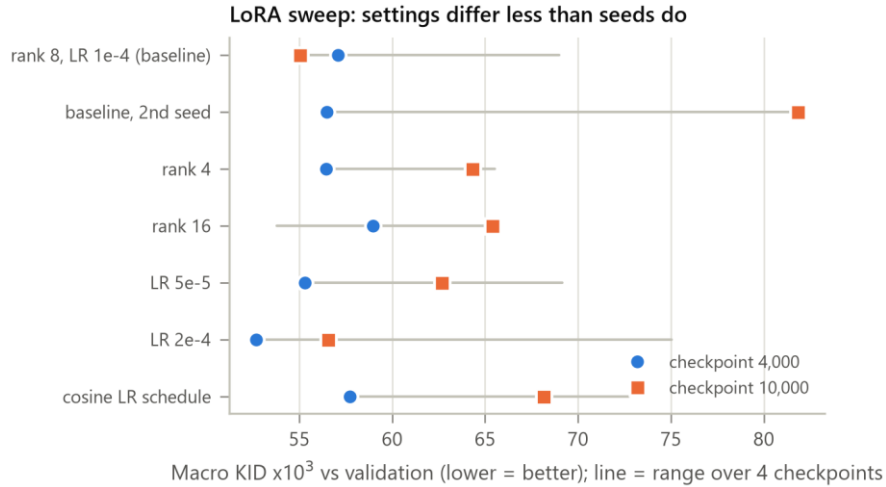


Fig. 7. Rank / learning-rate / schedule sweep: settings differ less than training seeds do.

TABLE V. Generator candidates on validation (30 steps, 100 images/class)

Candidate	Guidance	Macro KID $\times 10^3$	Macro FID	Class fidelity
seed 42 @ 10,000	1.5	56.6	74.7	0.436
seed 42 @ 10,000	2	55.0	72.3	0.550
seed 42 @ 10,000	3	57.5	72.5	0.702
seed 43 @ 4,000	1.5	60.4	81.0	0.312
seed 43 @ 4,000	2	56.5	76.8	0.397
seed 43 @ 4,000	3	57.8	76.2	0.546
seed 42 @ 4,000	1.5	56.9	76.3	0.324
seed 42 @ 4,000	2	57.1	74.7	0.424
seed 42 @ 4,000	3	66.8	79.2	0.568

Class fidelity: fraction of generated images a real-trained ResNet-18 assigns to the intended class (mean of 2 classifiers). Chosen: seed 42 @ 10,000 steps, guidance 3 (confirmed by the usefulness pilot).

Choosing guidance on validation: realism vs. class fidelity vs. usefulness (whiskers = range of 2 seeds)

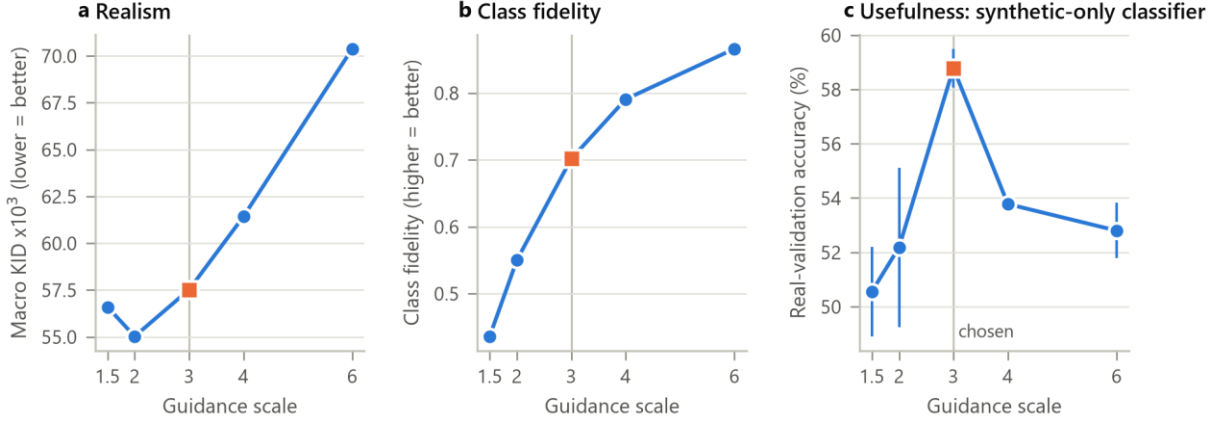


Fig. 8. Guidance pilot: realism (KID), class fidelity and usefulness (synthetic-only classifier on real validation images); guidance 3 is most useful.

Choosing sampling steps: more steps look more real but do not teach better (guidance 3)

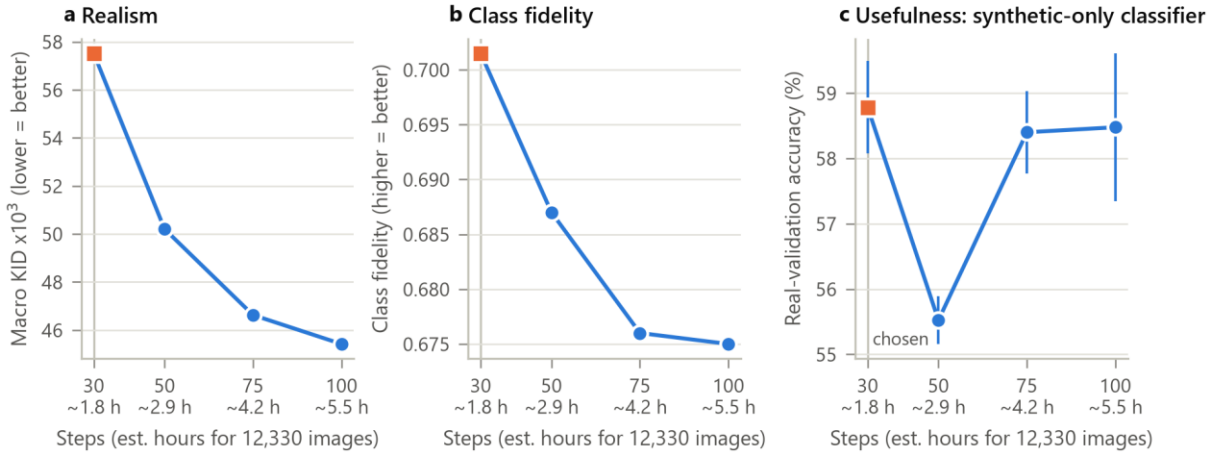


Fig. 9. Steps pilot: more steps give more realistic images (lower KID) but no more useful ones; 30 steps chosen.

TABLE VI. Sampling-setting pilots on validation (checkpoint 10,000; 1,000 images per setting)

Pilot	Guidance	Steps	Macro KID x1e3	Class fidelity	Synthetic-only val. acc % (2 seeds)	Macro-F1
guidance	1.5	30	56.6	0.436	50.6 (48.9-52.2)	0.465
guidance	2	30	55.0	0.550	52.2 (49.3-55.1)	0.498
guidance	3	30	57.5	0.702	58.8 (58.1-59.5)	0.554
guidance	4	30	61.4	0.790	53.8 (53.7-53.9)	0.502
guidance	6	30	70.4	0.866	52.8 (51.8-53.8)	0.496
steps	3	30	57.5	0.702	58.8 (58.1-59.5)	0.554
steps	3	50	50.2	0.687	55.5 (55.2-55.9)	0.517
steps	3	75	46.6	0.676	58.4 (57.8-59.0)	0.548
steps	3	100	45.4	0.675	58.5 (57.3-59.6)	0.551
control	real 100/class	-	-	-	74.4 (73.9-74.8)	0.728

Usefulness = ResNet-18 trained only on the 1,000 generated images, scored on real validation images. Chosen: guidance 3, 30 steps (fewest steps within 2 points of the best).

B. Final synthetic set: realism, class fidelity and memorization

TABLE VII. The final synthetic set (12,330 images): realism, class fidelity, memorization

Class	Images	FID synthetic	FID real ref.	KID x1e3	Class fidelity	Real val. recall	NN sim. synthetic (median)	NN sim. validation (median)	Above max real NN sim.
Disturbed	736	61.3	24.5	59.0	0.289	0.554	0.928	0.952	0
Merging	1287	63.4	19.4	69.5	0.870	0.903	0.928	0.952	0
Round Smooth	1844	41.9	9.4	53.6	0.909	0.964	0.944	0.962	0
In-between Round	1415	48.4	11.2	64.8	0.784	0.940	0.941	0.961	0
Cigar-Shaped Smooth	233	80.6	39.3	55.1	0.283	0.840	0.923	0.951	0
Barred Spiral	1427	67.7	20.1	65.1	0.804	0.875	0.915	0.943	0
Unbarred Tight Spiral	1273	53.4	19.2	50.3	0.788	0.761	0.930	0.950	0
Unbarred Loose Spiral	1816	52.6	14.7	55.5	0.669	0.798	0.924	0.947	0
Edge-on w/o Bulge	993	63.7	24.3	49.6	0.955	0.906	0.899	0.932	0
Edge-on w/ Bulge	1306	67.4	17.6	70.9	0.659	0.950	0.906	0.941	0

FID/KID vs validation at equal n. NN sim.: cosine similarity (Inception features) to the nearest working-pool image; 0 synthetic images are closer than the closest unseen validation image (no memorization).

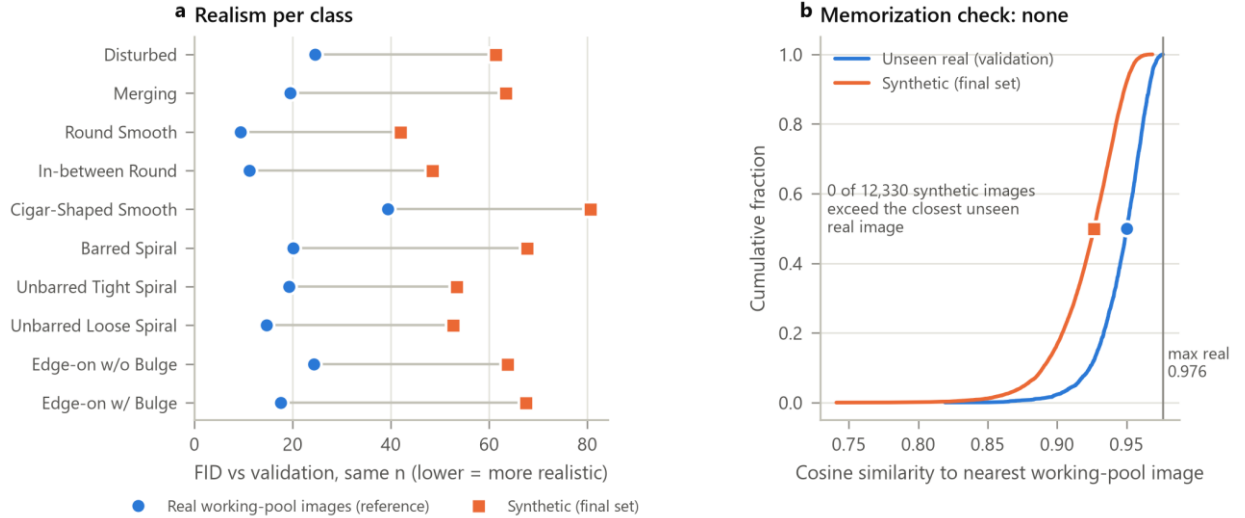


Fig. 10. (a) Per-class FID of the final set vs. the real reference; (b) nearest-neighbour similarity to the working pool: synthetic images are further from training images than unseen real ones (no memorization).

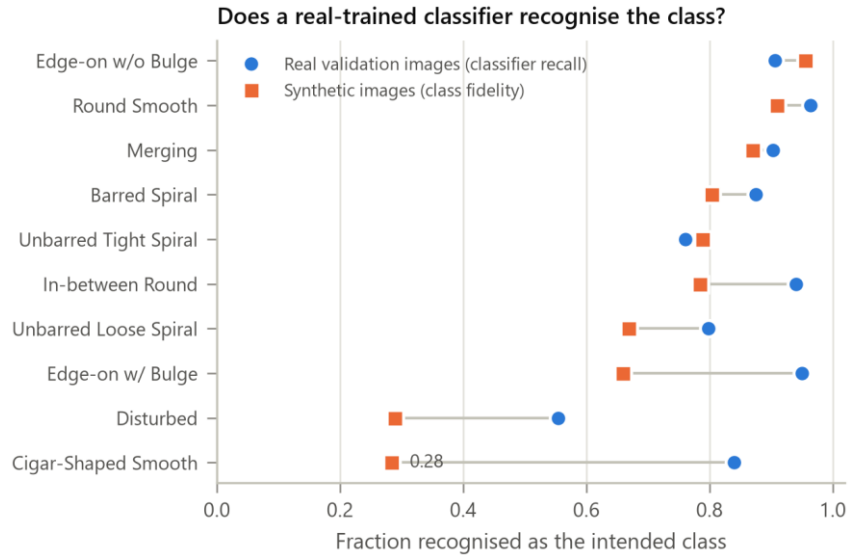


Fig. 11. Per class: how often a real-trained classifier recognizes synthetic vs. real images.

C. Per-class test results and comparative examples

Fig. 12 is the required comparative-examples figure: for ten held-out test galaxies (a seeded random draw fixed before viewing predictions) it gives the ground truth first, then the prediction of each model (real-only, synthetic-only, 50% mix; majority over seeds). In this draw the real-only and 50% models are right 9 of 10 times and synthetic-only 7 of 10. The patterns match Section VI: synthetic-only calls a Cigar-Shaped galaxy In-between Round, and the Disturbed galaxy fools all three. Ten images are an illustration, not a result.

Held-out test galaxies: true class vs. predictions of classifiers trained on real, synthetic and mixed data

Random draw fixed before viewing predictions (2 Cigar-Shaped Smooth + 1 from each of 8 other classes). Predictions: majority over seeds (agreeing seeds shown), from the saved one-time test evaluation.

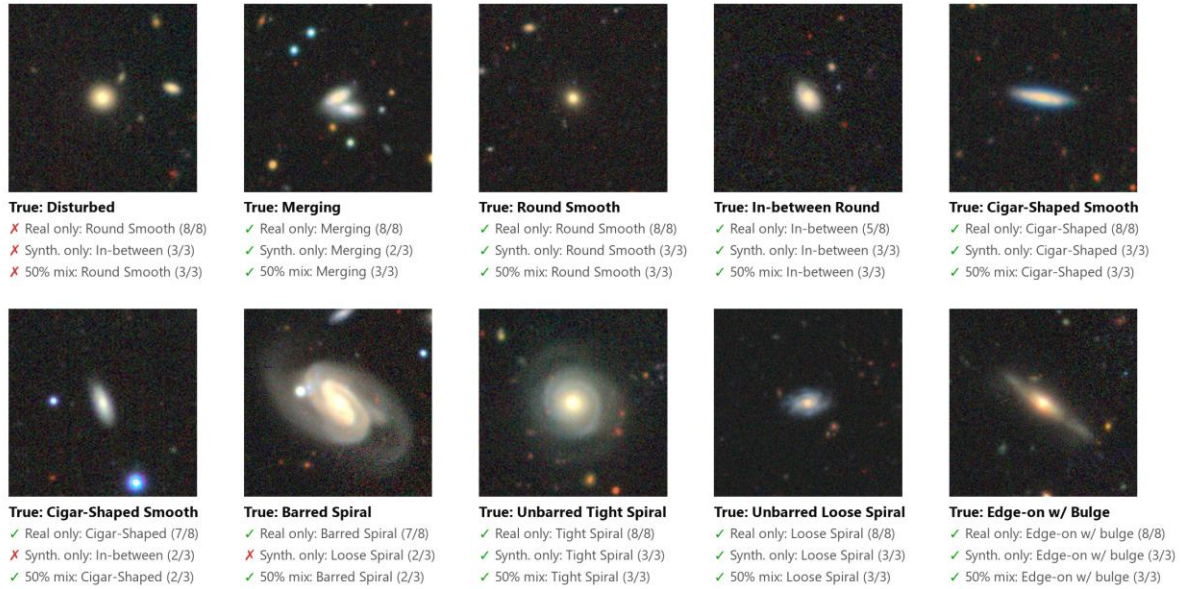


Fig. 12. Held-out test galaxies: true class and the predictions of the real-only, synthetic-only and 50%-mix classifiers (number of agreeing seeds in parentheses).

TABLE VIII. Replacing real with synthetic training data (total fixed at 12,330)

Synthetic share	Test accuracy % [95% CI]	Test macro-F1 [95% CI]	Delta acc. vs real-only (pts)	Val. accuracy %	Seeds
0%	86.6 [85.4, 87.6]	0.842 [0.827, 0.855]	-	86.2	8
10%	86.1 [84.9, 87.2]	0.840 [0.825, 0.853]	-0.44 [-0.96, +0.08]	85.8	3
25%	85.6 [84.4, 86.7]	0.832 [0.817, 0.846]	-0.94 [-1.51, -0.37]	85.3	3
50%	84.8 [83.6, 86.0]	0.824 [0.808, 0.838]	-1.73 [-2.38, -1.06]	84.0	3
75%	82.5 [81.3, 83.8]	0.796 [0.780, 0.811]	-4.02 [-4.87, -3.20]	81.9	3
90%	79.2 [77.8, 80.5]	0.762 [0.745, 0.777]	-7.36 [-8.41, -6.33]	78.5	3
100%	66.9 [65.3, 68.5]	0.621 [0.603, 0.638]	-19.61 [-21.19, -18.10]	66.4	3

Synthetic-only reaches 77.3% of real-only test accuracy (73.7% of macro-F1).

TABLE IX. Per-class test precision / recall / F1 (seed means)

Class	Real P	Real R	Real F1	50% P	50% R	50% F1	Synth. P	Synth. R	Synth. F1
Disturbed	0.650	0.488	0.557	0.601	0.465	0.524	0.312	0.285	0.287
Merging	0.878	0.897	0.887	0.852	0.879	0.865	0.740	0.598	0.660
Round Smooth	0.918	0.935	0.927	0.919	0.939	0.929	0.898	0.811	0.852
In-between Round	0.920	0.940	0.930	0.914	0.926	0.920	0.741	0.839	0.786
Cigar-Shaped Smooth	0.767	0.775	0.769	0.726	0.793	0.758	0.291	0.533	0.375
Barred Spiral	0.890	0.911	0.900	0.852	0.919	0.884	0.701	0.692	0.692
Unbarred Tight Spiral	0.776	0.835	0.804	0.762	0.807	0.784	0.664	0.510	0.573
Unbarred Loose Spiral	0.809	0.776	0.792	0.798	0.721	0.757	0.542	0.666	0.596
Edge-on w/o Bulge	0.895	0.918	0.906	0.890	0.894	0.892	0.861	0.493	0.625
Edge-on w/ Bulge	0.950	0.952	0.951	0.919	0.937	0.928	0.692	0.861	0.767

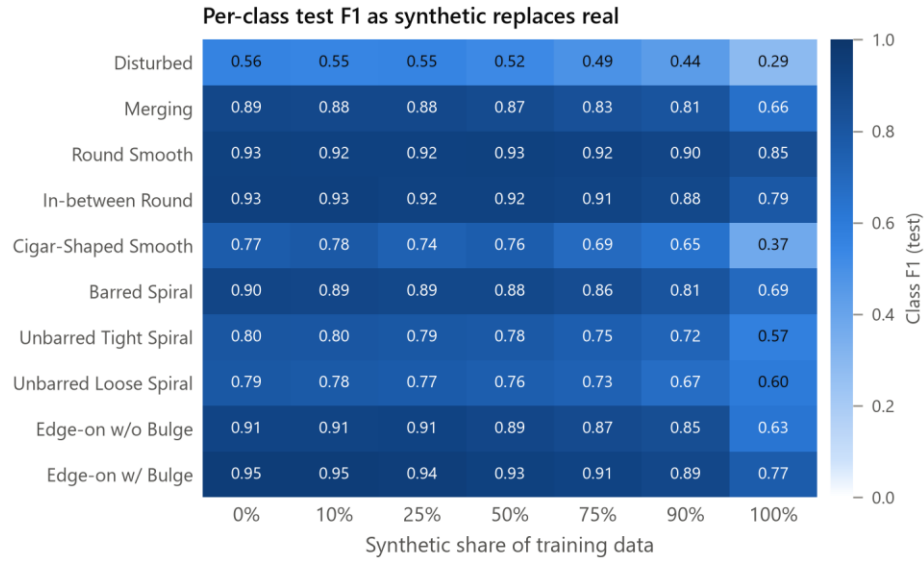


Fig. 13. Per-class test F1 across the replacement curve: Disturbed and Cigar-Shaped degrade most.

D. Augmentation and focus classes

TABLE X. Adding synthetic images on top of all real data

Training data	Test accuracy % [95% CI]	Test macro-F1 [95% CI]	Delta acc. vs real-only (pts)	Seeds
Real only (12,330)	86.6 [85.4, 87.6]	0.842 [0.827, 0.855]	-	8
+25% synthetic	86.5 [85.2, 87.6]	0.843 [0.828, 0.857]	-0.10 [-0.63, +0.45]	3
+50% synthetic	86.5 [85.3, 87.7]	0.840 [0.825, 0.854]	-0.04 [-0.61, +0.57]	3
+100% synthetic	86.8 [85.6, 88.0]	0.843 [0.827, 0.857]	+0.25 [-0.31, +0.84]	3

All differences include 0: no measurable effect.

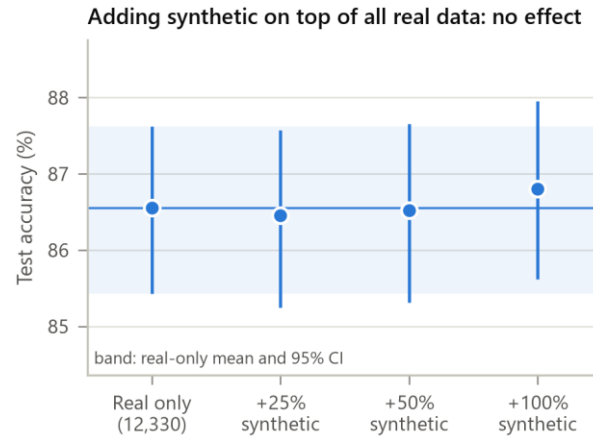
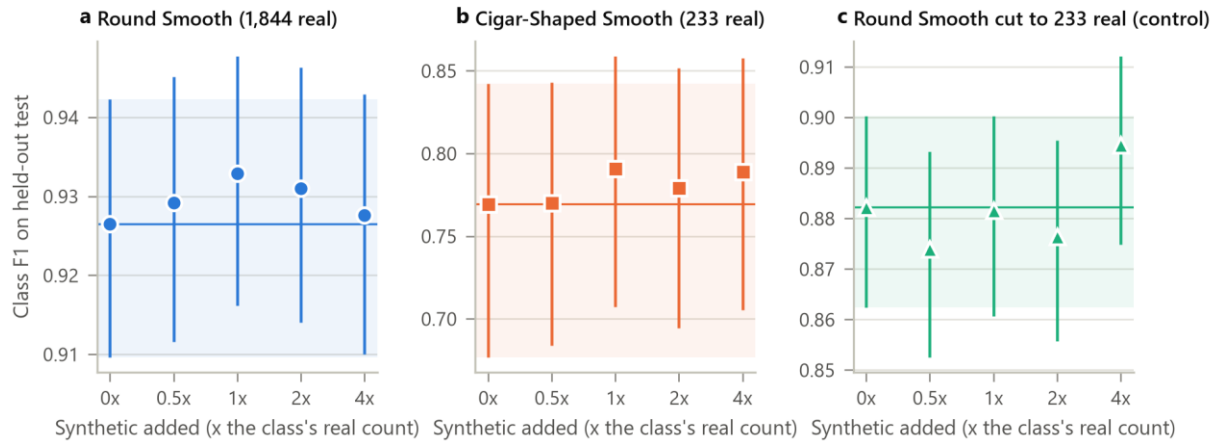


Fig. 14. Adding 25/50/100% synthetic images to all real data: no effect.

TABLE XI. Focus-class analysis: class F1 on the test set when adding synthetic images of that class

Class (real count)	Synthetic added	Class F1 [95% CI]	Delta F1 vs no synthetic [95% CI]	Overall acc. %	Seeds
Round Smooth (1,844 real)	0x	0.927 [0.910, 0.942]	-	86.6	8
Round Smooth (1,844 real)	0.5x	0.929 [0.912, 0.945]	+0.003 [-0.003, +0.009]	86.5	3
Round Smooth (1,844 real)	1x	0.933 [0.916, 0.948]	+0.006 [+0.001, +0.013]	86.6	3
Round Smooth (1,844 real)	2x	0.931 [0.914, 0.946]	+0.004 [-0.002, +0.011]	86.5	3
Round Smooth (1,844 real)	4x	0.928 [0.910, 0.943]	+0.001 [-0.005, +0.007]	86.7	3
Cigar-Shaped Smooth (233 real)	0x	0.769 [0.677, 0.842]	-	86.6	8
Cigar-Shaped Smooth (233 real)	0.5x	0.770 [0.684, 0.843]	+0.001 [-0.027, +0.028]	86.8	8
Cigar-Shaped Smooth (233 real)	1x	0.791 [0.707, 0.859]	+0.021 [-0.006, +0.050]	86.3	8
Cigar-Shaped Smooth (233 real)	2x	0.779 [0.695, 0.852]	+0.010 [-0.024, +0.044]	86.3	8
Cigar-Shaped Smooth (233 real)	4x	0.789 [0.705, 0.858]	+0.020 [-0.009, +0.048]	86.6	8
Round Smooth cut to 233 (control)	0x	0.882 [0.862, 0.900]	-	85.0	8
Round Smooth cut to 233 (control)	0.5x	0.874 [0.853, 0.893]	-0.008 [-0.016, -0.001]	84.9	8
Round Smooth cut to 233 (control)	1x	0.882 [0.861, 0.900]	-0.001 [-0.008, +0.006]	84.6	8
Round Smooth cut to 233 (control)	2x	0.876 [0.856, 0.895]	-0.006 [-0.013, +0.001]	84.9	8
Round Smooth cut to 233 (control)	4x	0.895 [0.875, 0.912]	+0.012 [+0.005, +0.019]	85.6	8

Rule fixed before testing: ~20 differences are tested, so isolated CIs excluding 0 are expected by chance; no consistent pattern in any class.



Whiskers: 95% bootstrap CI; band: the no-synthetic baseline's CI. 8 seeds for Cigar-Shaped and the control, 3 for Round Smooth. No consistent effect in any panel.

Fig. 15. Class F1 vs. synthetic ratio for Round Smooth, Cigar-Shaped Smooth and the scarcity control (Round Smooth cut to 233 real images): no consistent effect.

TABLE XII. What synthetic images add, by amount of real data (paired: identical real images)

Real data available	Real only: test acc. %	+ synthetic (to 12,330): test acc. %	Gain (pts) [95% CI]
10% (~1,233)	77.0 [75.7, 78.4]	79.2 [77.8, 80.5]	+2.16 [+1.19, +3.13]
25% (~3,082)	81.9 [80.6, 83.2]	82.5 [81.3, 83.8]	+0.60 [-0.18, +1.38]
50% (~6,165)	84.7 [83.4, 85.8]	84.8 [83.6, 86.0]	+0.17 [-0.50, +0.89]
75% (~9,248)	85.8 [84.6, 86.9]	85.6 [84.4, 86.7]	-0.15 [-0.83, +0.50]
90% (~11,097)	86.1 [84.9, 87.2]	86.1 [84.9, 87.2]	+0.01 [-0.60, +0.66]

The gain grows as real data gets scarcer and is clear only at 10% real.

E. Training curves and robustness check

Classifier training vs. validation loss and accuracy per epoch (seed means)

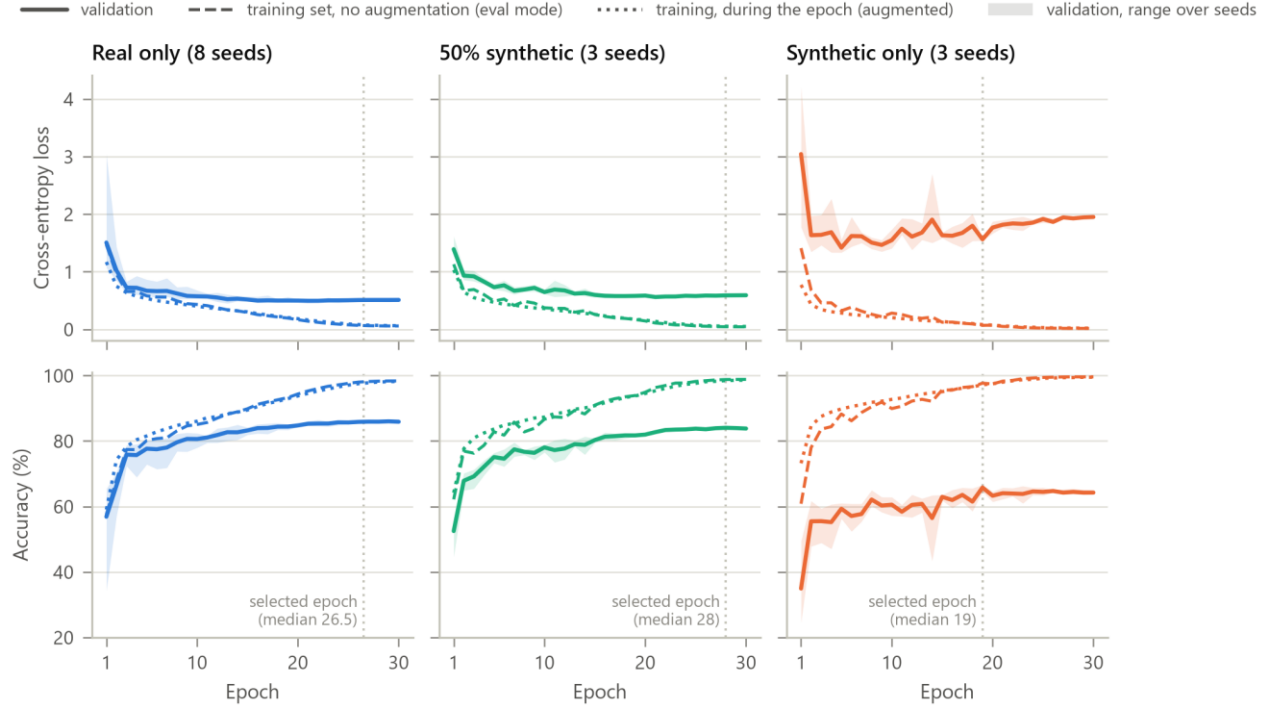


Fig. 16. Training vs. validation loss and accuracy per epoch for real-only, 50% and synthetic-only classifiers (seed means; band: seed range; selected epoch marked). Curves come from re-runs with extra logging, each verified bit-identical to the original run.

TABLE XIII. Robustness check (validation only): classifier training length

Classifier training	Real-only val. acc. %	Synthetic-only val. acc. %	Gap (pts)	Synthetic as % of real
30 epochs (used)	86.5 +/- 0.2	66.4 +/- 0.2	20.1	76.8%
50 epochs (check)	86.1 +/- 0.4	66.9 +/- 1.2	19.1	77.8%

Seeds 0-2 for both. 50 epochs changes the gap by -0.9 points: the conclusion is not sensitive to the schedule. The test set was not used.

F. ROC curves and AUC (validation set)

ROC curves per class, one-vs-rest, on the validation set (seed-mean curves)

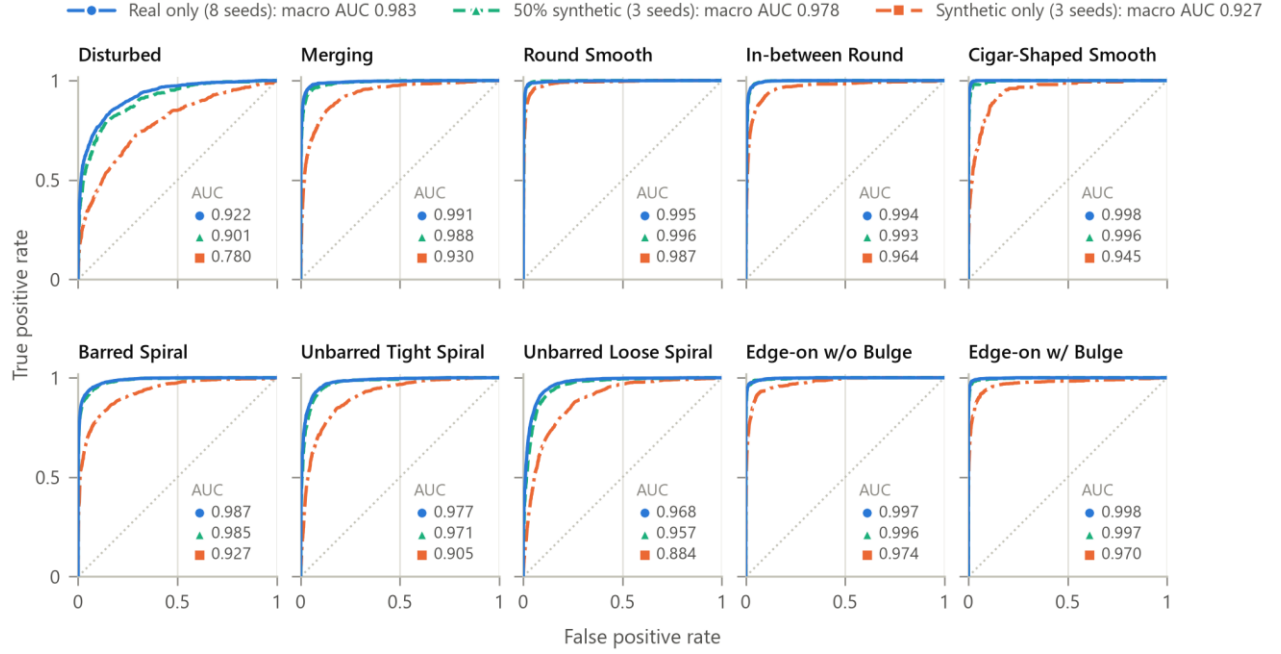


Fig. 17. One-vs-rest ROC curves per class (validation set) for real-only, 50% and synthetic-only classifiers, with per-class AUC.

TABLE XIV. One-vs-rest ROC AUC per class on the validation set

Class	Real only AUC [95% CI]	50% synthetic AUC [95% CI]	Synthetic only AUC [95% CI]	Synthetic only - real only	50% - real only
Disturbed	0.922 [0.900, 0.941]	0.901 [0.875, 0.923]	0.780 [0.741, 0.818]	-0.141 [-0.179, -0.104]	-0.021 [-0.035, -0.008]
Merging	0.991 [0.987, 0.995]	0.988 [0.983, 0.992]	0.930 [0.913, 0.945]	-0.062 [-0.078, -0.047]	-0.004 [-0.007, -0.000]
Round Smooth	0.995 [0.993, 0.997]	0.996 [0.994, 0.997]	0.987 [0.981, 0.991]	-0.009 [-0.014, -0.005]	+0.001 [-0.001, +0.002]
In-between Round	0.994 [0.991, 0.996]	0.993 [0.990, 0.995]	0.964 [0.952, 0.975]	-0.030 [-0.042, -0.019]	-0.001 [-0.002, +0.001]
Cigar-Shaped Smooth	0.998 [0.997, 0.999]	0.996 [0.993, 0.998]	0.945 [0.922, 0.966]	-0.053 [-0.076, -0.033]	-0.003 [-0.005, -0.001]
Barred Spiral	0.987 [0.982, 0.991]	0.985 [0.980, 0.989]	0.927 [0.913, 0.941]	-0.060 [-0.073, -0.048]	-0.003 [-0.006, +0.000]
Unbarred Tight Spiral	0.977 [0.969, 0.983]	0.971 [0.962, 0.979]	0.905 [0.886, 0.921]	-0.072 [-0.088, -0.057]	-0.005 [-0.008, -0.002]
Unbarred Loose Spiral	0.968 [0.961, 0.975]	0.957 [0.948, 0.965]	0.884 [0.868, 0.900]	-0.084 [-0.100, -0.070]	-0.011 [-0.016, -0.007]
Edge-on w/o Bulge	0.997 [0.995, 0.999]	0.996 [0.993, 0.998]	0.974 [0.964, 0.982]	-0.023 [-0.033, -0.015]	-0.001 [-0.002, +0.001]
Edge-on w/ Bulge	0.998 [0.997, 0.999]	0.997 [0.995, 0.999]	0.970 [0.958, 0.981]	-0.028 [-0.040, -0.018]	-0.001 [-0.003, +0.000]
Macro AUC	0.983 [0.980, 0.985]	0.978 [0.975, 0.981]	0.927 [0.920, 0.933]	-0.056 [-0.062, -0.050]	-0.005 [-0.007, -0.003]

Validation set (2,607 images); seed means (8 / 3 / 3 seeds); 95% paired bootstrap CIs over validation images (2,000 resamples). Macro AUC = mean of the 10 one-vs-rest AUCs. Computed on validation because the one-time test evaluation saved only predicted classes, not scores; the test set was not re-run.

G. Dataset details

TABLE XV. Dataset composition after cleaning (Galaxy10 DECaLS)

Class	Original	Exact duplicates removed	Overlapping cutouts removed	Final	Working pool (train)	Validation	Test
0 Disturbed	1081	30	0	1051	736	158	157
1 Merging	1853	14	48	1791	1287	253	251
2 Round Smooth	2645	10	4	2631	1844	391	396
3 In-between Round	2027	6	4	2017	1415	301	301
4 Cigar-Shaped Smooth	334	1	0	333	233	50	50
5 Barred Spiral	2043	5	2	2036	1427	304	305
6 Unbarred Tight Spiral	1829	10	1	1818	1273	272	273
7 Unbarred Loose Spiral	2628	33	6	2589	1816	387	386
8 Edge-on w/o Bulge	1423	4	2	1417	993	213	211
9 Edge-on w/ Bulge	1873	8	2	1863	1306	278	279
Total	17736	121	69	17546	12330	2607	2609

Exact duplicates: pixel-identical images with conflicting labels, removed before the stratified 70/15/15 split. Overlapping cutouts: validation/test images covering $\geq 25\%$ of the same sky as an image in another split, removed after splitting (working pool unchanged).

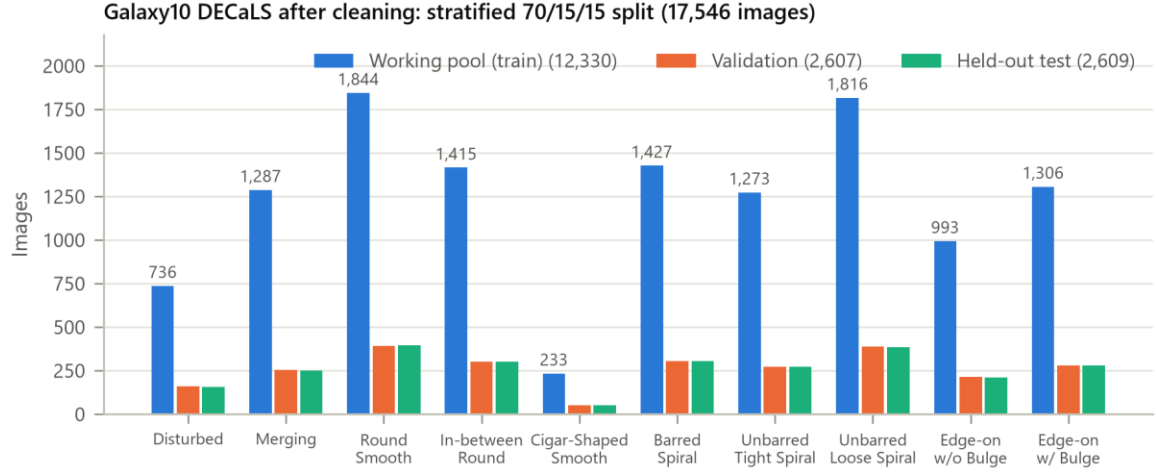


Fig. 18. Per-class counts of the working pool, validation and test sets after cleaning (17,546 images).

Galaxy10 DECaLS: real examples of each class (training split, random draw)

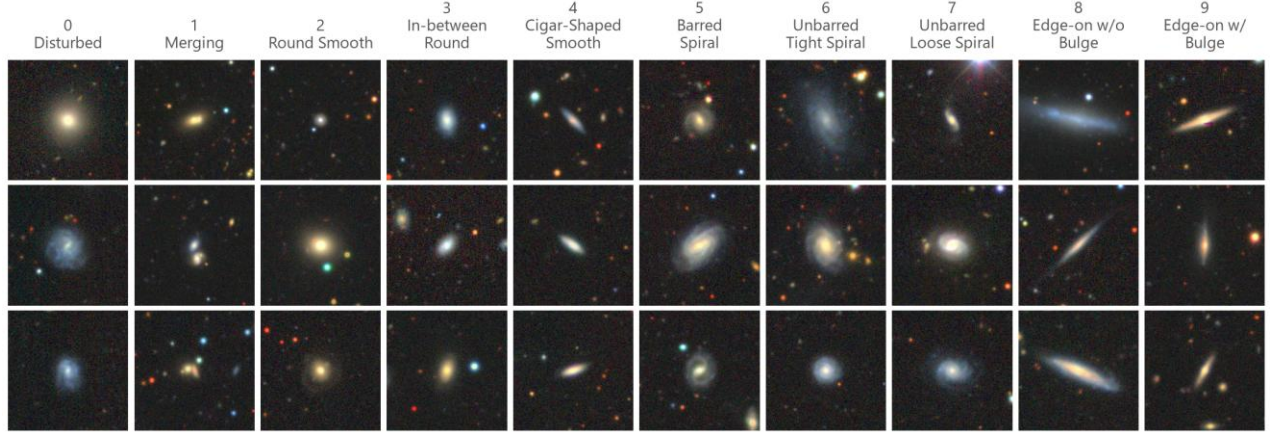


Fig. 19. Three real working-pool examples per class (seeded random draw).

H. Full settings, model cost and inference benchmark

TABLE XVI. Model size, training time and inference cost (one NVIDIA RTX 5090)

Model	Parameters (total / trained)	Training	Inference, ms / image	GFLOPs / image	Peak GPU memory
Generator: SD 1.5 U-Net + VAE + LoRA + class table	945.4 M / 2.24 M	12,330 real images, 10,000 steps, 60 min	920 (batch 1); 479 (batch 20, as used)	50,848 (30 guided steps $\times$ 1,611 + VAE decode 2,515)	10.7 GiB (training); 12.1 GiB (sampling)
Classifier: ResNet-18	11.18 M / 11.18 M	12,330 images, 30 epochs, 1.5 min	1.60 (GPU forward); 1.74 (end-to-end), batch 1	4.74	not measured

Generator: guidance 3, 30 DPM-Solver++ [22] steps, fp16, 512 px including VAE decode; the CLIP text encoder (123.1 M) is not used at sampling. Generating the 12,330-image set took 99.5 min. FLOPs: 2 per multiply-accumulate (PyTorch FlopCounterMode). Hardware: AMD Ryzen 7 9800X3D, 32 GB RAM, Windows 11, PyTorch 2.14, CUDA 13.0.

TABLE XVII. Model and training settings

Stage	Setting	Value
Diffusion	Base model	runwayml/stable-diffusion-v1-5 (Stable Diffusion 1.5)
Diffusion	Fine-tuning	LoRA rank 8, alpha 8 on to_q/to_k/to_v/to_out.0
Diffusion	Conditioning	learned 11 x 77 x 768 class table (10 classes + null), CLIP-initialised
Diffusion	Optimiser	AdamW, LR 0.0001, weight decay 0.01, constant_with_warmup (500 warmup)
Diffusion	Training	10,000 steps (= 6.5 epochs of the 12,330 training images), batch 8, 512px, bf16, cond. dropout 0.1, flips + 90-deg rotations, seed 42
Diffusion	Parameters	U-Net 859,520,964 (frozen) + LoRA 1,594,368 on 128 attention projections (trained) + class table 650,496 (trained); VAE 83,653,863 (frozen); CLIP text encoder 123,060,480 (used once, to initialise the class table)
Diffusion	Training time	60 min for 10,000 steps (0.36 s/step)
Diffusion	Selected checkpoint	10,000 steps (chosen on validation KID + class fidelity)
Sampling	Sampler	DPM-Solver++ (2nd order), 30 steps, classifier-free guidance 3.0
Sampling	Output	512 px generated, Lanczos-downsampled to native 256 px
Sampling	Generation time	99.5 min for the 12,330-image synthetic set (0.48 s/image, batches of 20)
Classifier	Model	ResNet-18, ImageNet-pretrained, 256 px input
Classifier	Parameters	11,181,642 (all trained; ImageNet ResNet-18 with a new 10-class output layer)
Classifier	Optimiser	AdamW, LR 0.001 (tuned on validation from 1e-4 / 3e-4 / 1e-3), weight decay 0.05, 1 warmup epoch + cosine
Classifier	Training	30 epochs, batch 128, bf16, flips + 90-deg rotations, best epoch by validation accuracy
Classifier	Seeds	3 per condition (8 for real-only, Cigar-Shaped and scarcity-control conditions)
Classifier	Training time	Real only 1.5 min; 50% synthetic 1.5 min; Synthetic only 1.5 min per run (30 epochs incl. per-epoch validation; mean over seeds)
Evaluation	Realism	per-class FID / KID (Inception-v3 pool features) vs validation
Evaluation	Classification	accuracy, macro-F1, per-class P/R/F1 on the held-out test set (used once); 95% paired bootstrap CIs, 2,000 resamples
Hardware	All training and inference	1x NVIDIA GeForce RTX 5090 (32 GB); AMD Ryzen 7 9800X3D 8-Core Processor; 31 GB RAM; Windows 11; PyTorch 2.14.0+cu130, CUDA 13.0

TABLE XVIII. Inference speed and compute on one NVIDIA GeForce RTX 5090 (per image)

Model	Mode	Mean ms	Median ms	p95 ms	Images/s	Timed images	GFLOPs
Diffusion generator	batch 1 (latency), per image	920.2	899.0	1013.2	1.09	20	50,847.7
Diffusion generator	batch 20 (as used), per image	479.2	479.2	480.5	2.09	100	50,847.7
- U-Net + LoRA	one denoising step (guidance: 2 evaluations)	-	-	-	-	-	1,611.1
- VAE decoder	one 512 px image	-	-	-	-	-	2,514.5
ResNet-18 classifier	batch 1, GPU forward only	1.60	1.56	1.91	624	1000	4.7
ResNet-18 classifier	batch 1, end-to-end	1.74	1.68	2.09	575	1000	4.7

Generator: checkpoint-10000, guidance 3, 30 DPM-Solver++ steps, fp16, 512 px incl. VAE decode (+ 2.1 ms CPU resize to 256 px, not included); one image = 30 guided U-Net steps + one VAE decode; one U-Net evaluation = 805.6 GFLOPs. Classifier: ResNet-18 at 256 px, bf16; end-to-end = image upload + normalisation + forward + softmax download. Warm-up excluded; GPU synchronised around every timed call. FLOPs: PyTorch FlopCounterMode, 2 FLOPs per multiply-accumulate, matrix multiplications, convolutions and attention (analysis/model_stats.py); '-' = not timed separately.

I. Pseudocode

Algorithm 1. Experiment protocol, as run (Fig. 1).

```

1:  function RUNBENCHMARK( $D, S$ )
    ▷  $D$ : labelled galaxy images;  $S$ : classifier seeds
    ▷ Output: test metrics with paired 95% CIs for every condition
2:   $D \leftarrow D$  without pixel-identical duplicates that carry conflicting labels
3:   $(P, V, T) \leftarrow \text{STRATIFIEDSPLIT}(D, 70/15/15, \text{seed } 42)$ 
4:  Remove from  $V$  and  $T$  every image sharing  $\geq 25\%$  of its sky with another split
5:  for each LoRA setting  $\theta$  in the sweep do train  $G_\theta$  on  $P$ ; score checkpoints on  $V$ 
6:   $G \leftarrow$  best checkpoint on  $V$  (KID, class fidelity)
7:  Choose guidance  $w$  and steps  $K$  by usefulness: classifier trained on  $G$ 's images, scored on  $V$ 
8:   $Y \leftarrow G(\text{per-class counts of } P; w, K)$  ▷ memorization check against  $P$ 
9:  for each condition  $c$  in  $A \cup B \cup C$  (incl. paired controls), seed  $s$  in  $S$  do
10:    $X \leftarrow \text{MIX}(P, Y, c, s)$  ▷ which real and synthetic images: drawn with  $s$ 
11:    $f_{c,s} \leftarrow$  train ResNet-18 on  $X$ ; keep the epoch with the best accuracy on  $V$ 
12:  end for
13:  Fix the analysis plan; evaluate every  $f_{c,s}$  on  $T$  once
14:  return accuracy, macro-F1, per-class P/R/F1, confusion matrices, paired bootstrap CIs
15: end function

```